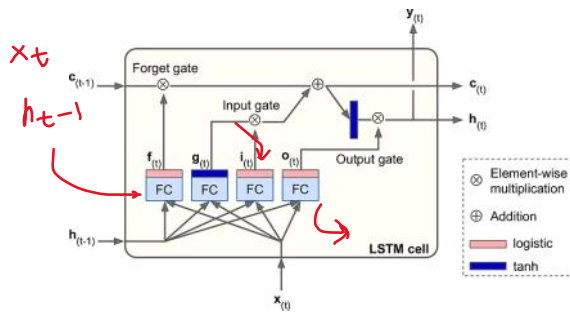
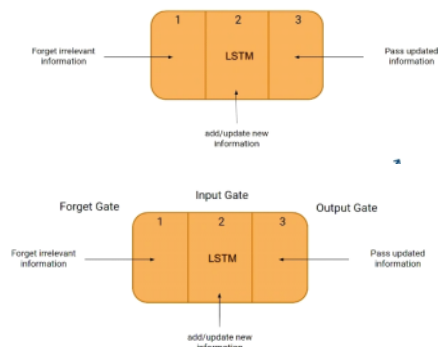
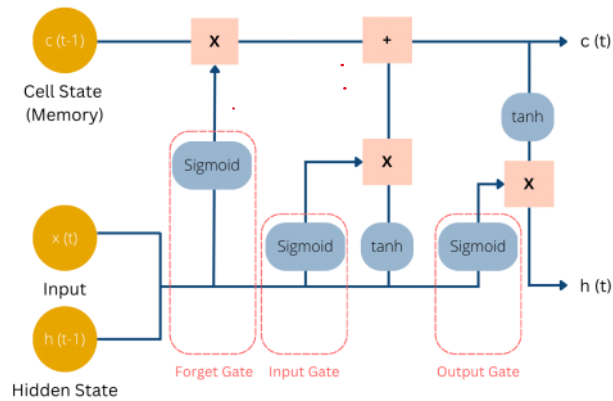




$$\begin{aligned} \rightarrow f(t) &= \sigma(w_f \cdot [h_{t-1}, x_t] + b_f) \\ \rightarrow i(t) &= \sigma(w_i \cdot [h_{t-1}, x_t] + b_i) \sim \underline{c_t} = \underline{\tanh}(w_c \cdot [h_{t-1}, x_t] + b_c) \\ \underline{o_t} &= \sigma(w_o \cdot [h_{t-1}, x_t] + b_o) h_t = a_t \times \tanh(c_t) \end{aligned}$$





Bob is a nice person. Dan on the other hand is evil

$$f_t = \sigma(x_t \cdot u_f + H_{t-1} \cdot w_f) = 0$$

Bob knows swimming. He told me over the phone that he had served the navy for four long years

$$i_t = \sigma(x_t \cdot u_i + H_{t-1} \cdot w_i)$$

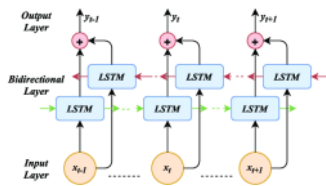
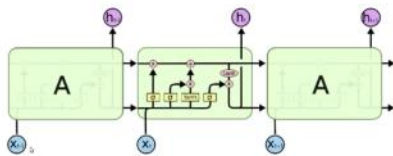
$$N_t = \tanh(x_t \cdot u_c + H_{t-1} \cdot w_c)$$

Bob single handedly fought the army and died for his country. For his contribution, brave __

$$O_t = \sigma(x_t \cdot u_o + H_{t-1} \cdot w_o)$$

$$H_t = O_t \tanh(H_t)$$

$$O_t = \text{Softmax}(H_t)$$



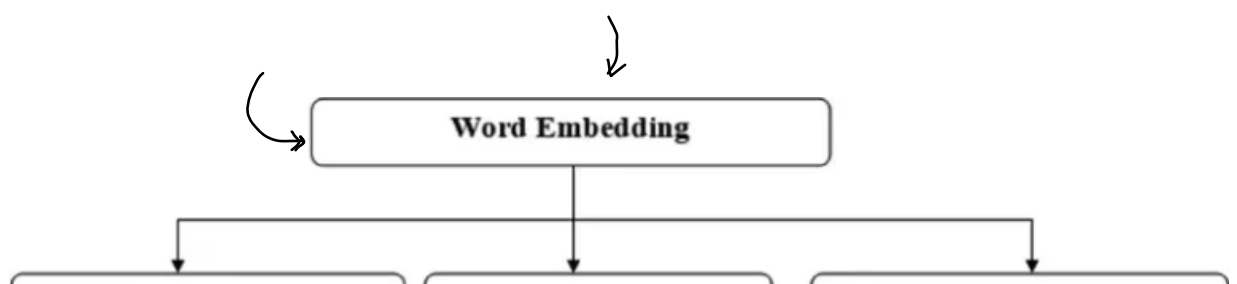
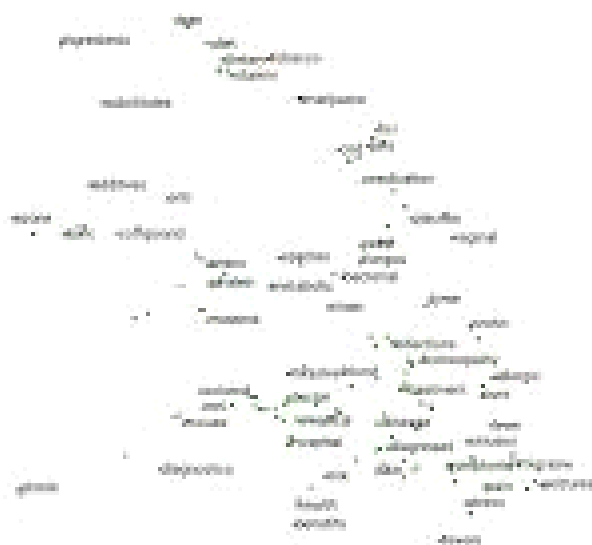
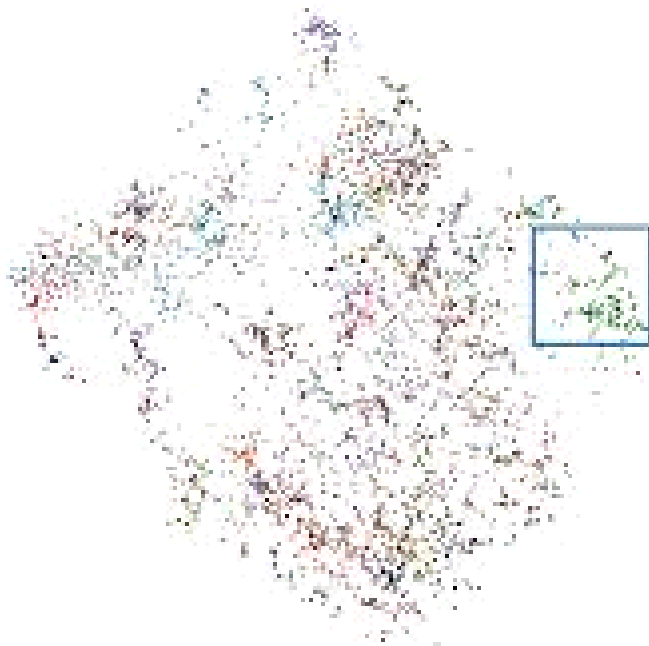
$$r = x_{h_{t-1}}$$

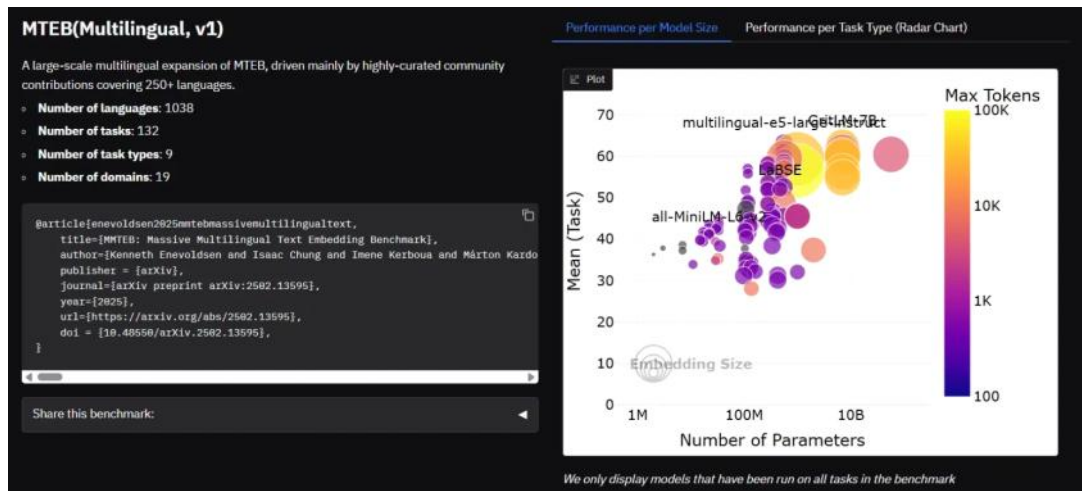
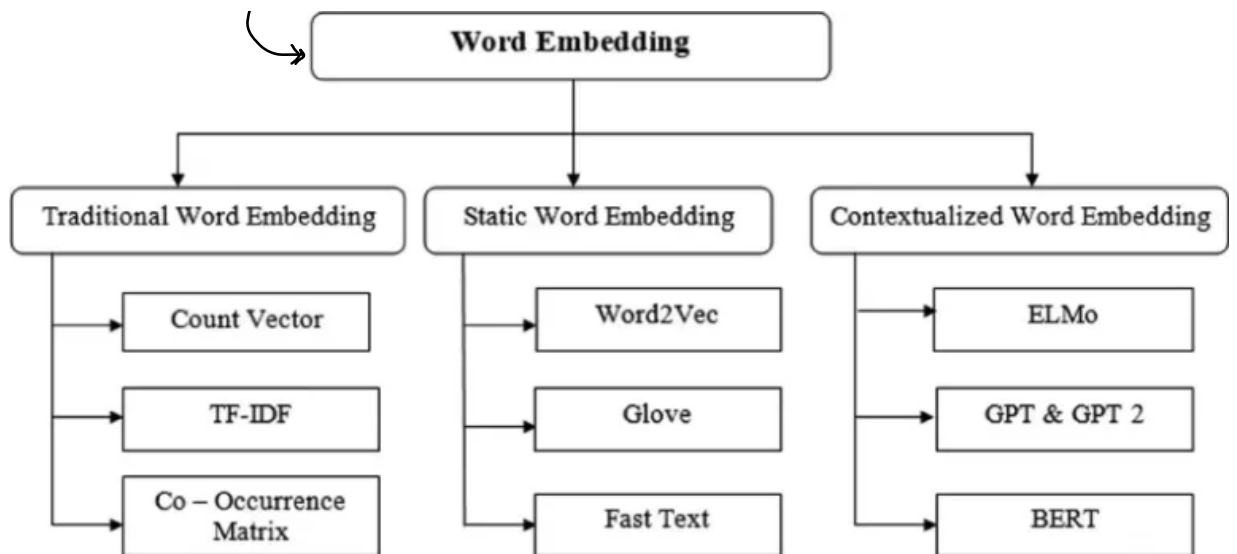
$$f_t = \sigma(w_f \times [h_{t-1}, x_t])$$

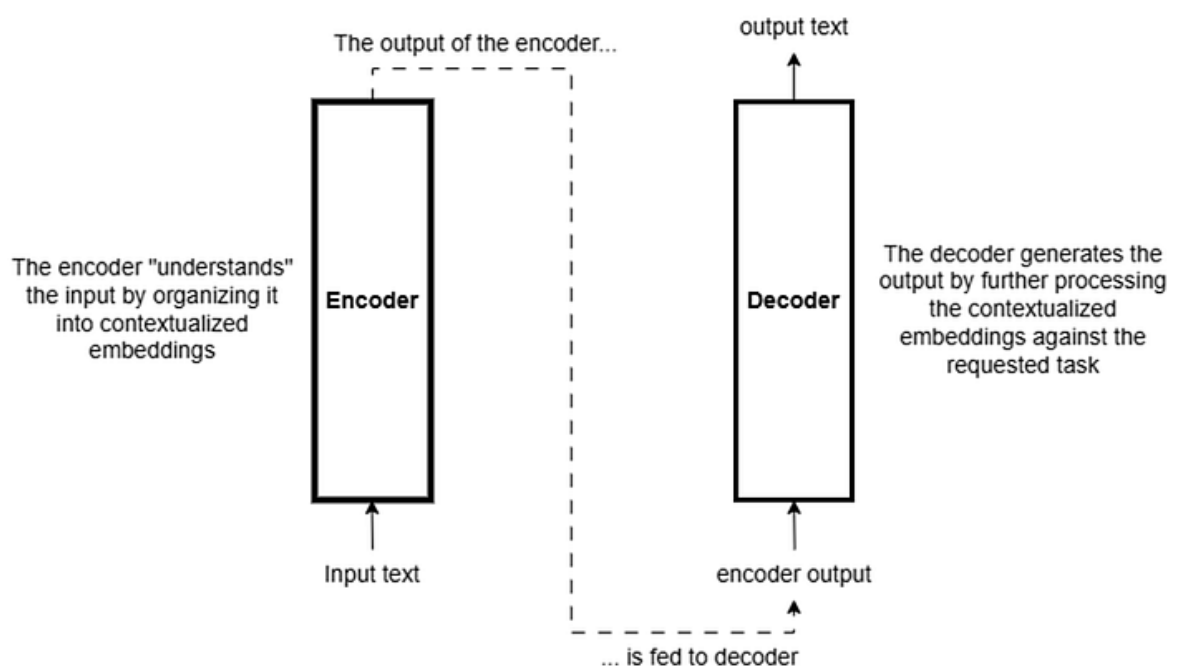
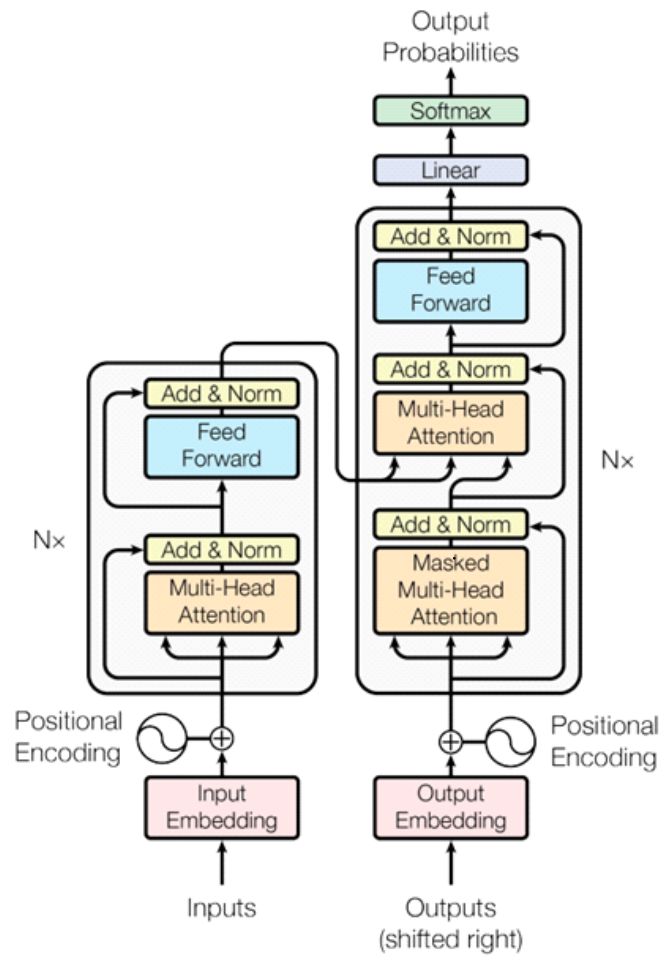
$$z_t = \sigma(w_z \times [h_{t-1}, x_t])$$

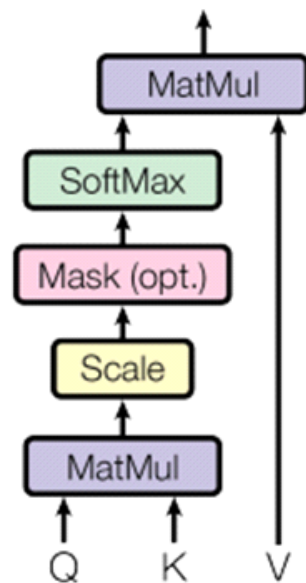
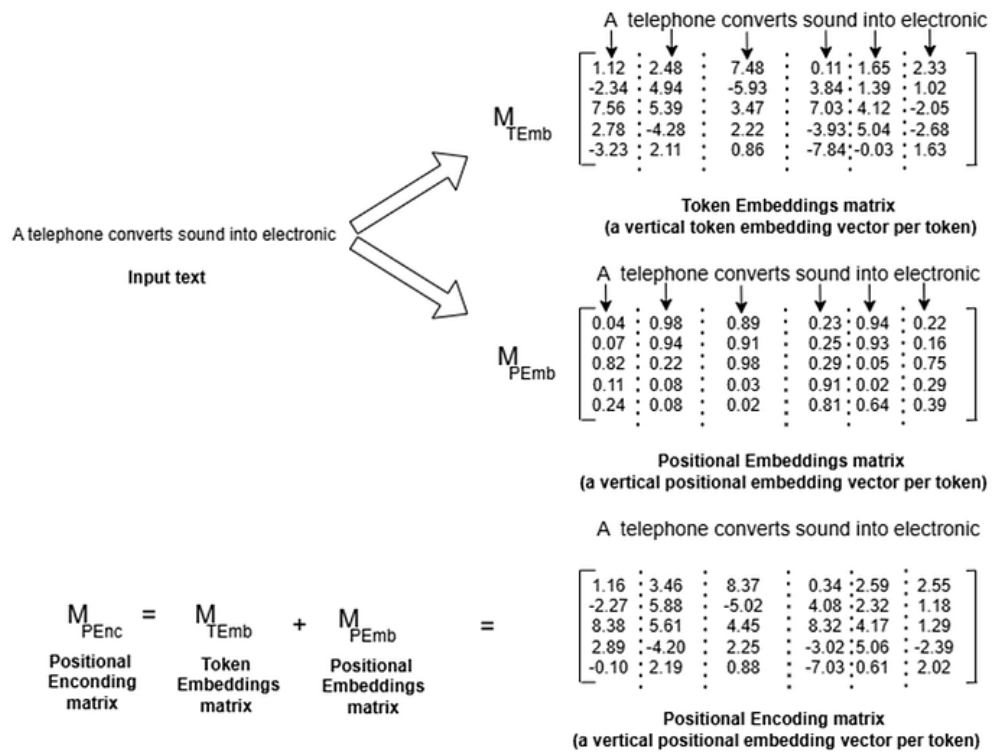
$$h_t = \tanh(w_h \times (f_t \times h_{t-1} + x_t))$$

$$\in [-1, +1]$$









Multi-Head Attention

